

Minimodule 4. Induction and recurrence

List of exercises (from 7th edition of the book)

Section 5.2 Exercises 12

Section 5.3 Exercises 2, 5a,b,c,d

Section 5.1 Exercises 5, 23, 40

12. Use strong induction to show that every positive integer n can be written as a sum of distinct powers of two, that is, as a sum of a subset of the integers $2^0 = 1, 2^1 = 2, 2^2 = 4$, and so on. [Hint: For the inductive step, separately consider the case where $k + 1$ is even and where it is odd. When it is even, note that $(k + 1)/2$ is an integer.]
2. Find $f(1)$, $f(2)$, $f(3)$, $f(4)$, and $f(5)$ if $f(n)$ is defined recursively by $f(0) = 3$ and for $n = 0, 1, 2, \dots$
- a) $f(n + 1) = -2f(n)$.
 - b) $f(n + 1) = 3f(n) + 7$.
 - c) $f(n + 1) = f(n)^2 - 2f(n) - 2$.
 - d) $f(n + 1) = 3^{f(n)/3}$.
5. Determine whether each of these proposed definitions is a valid recursive definition of a function f from the set of nonnegative integers to the set of integers. If f is well defined, find a formula for $f(n)$ when n is a nonnegative integer and prove that your formula is valid.
- a) $f(0) = 0, f(n) = 2f(n - 2)$ for $n \geq 1$
 - b) $f(0) = 1, f(n) = f(n - 1) - 1$ for $n \geq 1$
 - c) $f(0) = 2, f(1) = 3, f(n) = f(n - 1) - 1$ for $n \geq 2$
 - d) $f(0) = 1, f(1) = 2, f(n) = 2f(n - 2)$ for $n \geq 2$
5. Prove that $1^2 + 3^2 + 5^2 + \dots + (2n + 1)^2 = (n + 1)(2n + 1)(2n + 3)/3$ whenever n is a nonnegative integer.
23. For which nonnegative integers n is $2n + 3 \leq 2^n$? Prove your answer.
40. Prove that if $A_1, A_2, \dots, A_n$ and B are sets, then
- $$(A_1 \cap A_2 \cap \dots \cap A_n) \cup B = (A_1 \cup B) \cap (A_2 \cup B) \cap \dots \cap (A_n \cup B).$$